

Customer discovery

Part I · What are customers called where?

Sail · LakeCat · Pinax · MCP



One question across the company

“Sales says customer. Finance says account. Product says user. Where is each recorded?”

Find the table, field, definition, and responsible steward.

Three tables in the lakehouse

Department	Physical table	Fields
Sales	crm_customers	customer_id
Finance	billing_accounts	account_id, customer_ref
Product	product_users	user_id, customer_ref

Real Iceberg metadata and Parquet files · LakeCat catalog

What Pinax records

- A stable concept ID, definition, and alternative names
- Source references, steward, and review evidence
- Table bindings and stable field IDs at a registry revision

Concept approval and binding approval are separate.

Customer is an ambiguous name

Term	Meaning in this department
CRM customer	Organization with a commercial relationship
Billing account	Billing relationship; several per customer
Product user	Person with a login; may be a free user

Pinax returns the alternatives with their definitions.

Shared meaning at the field level

`crm_customers.customer_id`

`billing_accounts.customer_ref`

`product_users.customer_ref`

These three fields bind to **Enterprise customer identifier**.

`account_id` and `user_id` retain distinct concepts.

The agent asks through MCP

Discover tools

Search customer → account → user

Inspect every returned definition and binding

Find the enterprise customer reference

Console step 4 · Live signed metadata discovery

The answer names places and meanings

Sales: **customer** in `crm_customers`.

Finance: **account** in `billing_accounts`.

Product: **user** in `product_users`.

Follow-up: customer organizations, billing accounts, or individual users?

Maintain the mapping

1. Inventory and register tables and fields
2. Import draft concepts and bindings
3. Stewards review meanings and ambiguous names
4. Publish a revision and update consumer pins

Coverage means the authorized registered inventory.

Run the discovery demo

[Open demo.rust.ai](https://demo.rust.ai) · Iceberg

Choose the company table format in the header.

Four steps answer the customer naming question.

Part I ends here. Open Part II separately for graph workflows.

Part II · Graphs and workflows

An optional extension after catalog discovery

Grust graph storage, semantic bundles, and specialist orchestration.

Grust connects the data

Datasets → files → fields → concepts

Live demonstration

Result

**Dataset
descriptions**

Two Dataverse fixtures

Property graph

34 nodes and 33 edges in Grust's Delta tables through Sail

Verification

Dataset node read back from Sail

Provenance

OpenLineage appended to the lakehouse

Console step 8 · Build graph & run workflow

Marciana and semantic workflows

QueryGraph composes semantic and cognition capabilities into agent workflows.

- Navigator packages descriptions for discovery and reuse
- Specialists contribute domain summaries
- A supervisor coordinates and a synthesis agent combines
- TypeSec identities and lineage accompany the work

Console steps 9–10 · Semantic bundle and specialist workflow

Meaning travels with results

Surface	What an agent can reuse
Croissant	Record sets, fields, and bounded inline results
CDIF projection	Data elements linked to shared concepts
SKOS vocabulary	Labels, aliases, definitions, relationships
Lineage	The operations and inputs behind an answer

One central ontology feeds the semantic projections.